FORMULA 1: Race Intelligence Dashboard

OFFICIAL PROJECT DOCUMENTATION & ANALYSIS REPORT (2010–2026)

1. Problem Statement & Business Objective

Technical teams, race strategists, and motorsport analysts struggle to extract actionable insights from fragmented multi-season Formula 1 data. The lack of a centralized, interactive system for tracking telemetry, historical mechanical failures (Did Not Finish - DNF metrics), and driver/constructor trajectories limits real-time strategic decision-making and risk mitigation. This project aims to design and deploy an end-to-end Race Intelligence Dashboard that consolidates 16 years of F1 data into a live, interactive web application to uncover performance dynamics and reliability patterns.

2. Data Collection & Preparation

Source Data: The primary dataset utilized is `F1_Race_Intelligence_2010_2026.csv`, encompassing multi-season race results, global circuit locations, driver demographics, constructor points, and DNF classifications.

Data Cleaning & Transformation:

- **Geographic Normalization:** Standardized circuit locations and driver origin cities to ensure accurate spatial mapping across global coordinates.
- **Categorical Grouping:** Consolidated highly specific telemetry and DNF text fields into standardized operational categories (e.g., categorizing mechanical failures into *Engine, Gearbox, Hydraulics* and driver incidents into *Collisions, Spun Off*).
- **Data Integrity Checks:** Filtered out null telemetry records from incomplete historical races and validated the sequential logic of season-by-season championship point aggregations.

3. Technology Stack & Tools

PHASE	TOOL / TECHNOLOGY	PURPOSE
Data Visualization	Tableau Public	Dashboard design, dynamic cross-filtering, and 9-scene narrative story construction.
Backend Framework	Python (Flask)	Local web server routing and request handling.
Frontend Rendering	HTML5 / CSS	Web page structure and iframe embedding of the Tableau dashboard.
Environment	VS Code / Windows CLI	Code editing, directory management, and local server deployment.

4. Visualizations & Dashboard Architecture

The visual analytics layer was developed in Tableau and structured into a **5-chart interactive dashboard grid** and a **9-scene narrative story**.

- **Global Circuit Distribution:** A geographic map plotting the international footprint of F1 venues and race clustering across the 2010–2026 seasons.
- **Driver Reliability & DNF Diagnostics:** Stacked bar charts and heatmaps isolating technical failure frequencies and collision rates by constructor, allowing teams to audit historical mechanical risks.
- **Championship Trajectories:** Area and line charts tracking the accumulation of constructor points over time, visually mapping dynastic dominance (e.g., Mercedes' hybrid era, Red Bull's aero dominance).
- **Interactive Parameters:** Global `Season` and `DNF Reason` filters were implemented across the dashboard to enable seamless cross-filtering and granular data exploration.

5. Key Findings & Results

1. **Mechanical Vulnerabilities:** Early regulatory shifts (such as the 2014 V6 turbo-hybrid introduction) showed a distinct spike in specific DNF reasons, predominantly related to Power Unit (PU) and Energy Recovery System (ERS) failures.
2. **Constructor Dominance Patterns:** Visual trajectory mapping clearly delineated periods of constructor dominance, showing how early-season reliability directly correlates with late-season championship securing.
3. **Venue-Specific Incident Rates:** The cross-filtering revealed that specific circuits historically yield higher collision-based DNFs, providing predictive value for race strategists regarding safety car probabilities.

6. Web Application Integration & Deployment

To make the dashboard accessible outside of the Tableau ecosystem, the project was deployed as a standalone local web application:

1. **Flask Initialization:** A Python virtual environment was established, and a `Flask` application object was initialized in `app.py`.
2. **Route Configuration:** The root URL (`'/'`) was configured to execute the `render_template` function, serving the frontend.
3. **Tableau Embedding:** The Tableau Public HTML embed code was integrated into `index.html`, inheriting all native Tableau interactivity, tooltips, and cross-filtering.
4. **Local Hosting:** The application was successfully compiled and launched on a local WSGI development server (`http://127.0.0.1:5000`), proving the end-to-end integration of data analytics and web deployment.